
Quantum Computation and Quantum Information (EE798V)

Mid Semester Examination
Indian Institute of Technology Kanpur

Date: 25th February 2024

Time: 2 Hours

Total Marks: 50

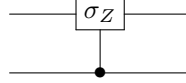
Note: Please read the instructions stated below.

- (a) All questions are mandatory.
- (b) Answers to all parts of a question must be written together.
- (c) For numerical questions, write approximations up to three decimal places.
- (d) Symbols carry their usual meanings.

Q. 1: Answer the following questions.

[1 × 10]

- (a) Argue whether $\lambda = 1 + i$ can be a valid eigenvalue of a unitary matrix.
- (b) What is the dimension of the Hilbert space that contains the states of a 4-qubit composite system?
- (c) Consider a 10-qubit composite system. How many different outcomes can be observed if a partial measurement is made on 3 arbitrarily chosen qubits?
- (d) Argue whether $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ is an entangled state.
- (e) Write down the unitary matrix describing the transformation induced by the following circuit.



- (f) Adjoint of an unitary operator, U is denoted as $U^\dagger$, and it is defined such that $\langle\phi|U\psi\rangle = \langle U^\dagger\phi|\psi\rangle$ for any $|\phi\rangle, |\psi\rangle$. Using this definition, show that $(UV)^\dagger = V^\dagger U^\dagger$.
- (g) Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be such that the cardinality of the set $\{x \in \{0, 1\}^n | f(x) = 1\}$ is 2^{n-2} . Note that the function is neither constant nor balanced. What is the probability that the function f will be classified as a constant function if the Deutsch-Jozsa protocol is applied to its corresponding quantum oracle U_f ?
- (h) Let $f : \{0, 1\}^2 \rightarrow \{0, 1\}$ be such that $f(0, 0) = f(0, 1) = f(1, 0) = 0$ and $f(1, 1) = 1$. Design a quantum oracle, U_f using the gates discussed in the class such that $U_f|x\rangle|b\rangle = |x\rangle|b \oplus f(x)\rangle$, where $x \in \{0, 1\}^2$, and $b \in \{0, 1\}$.
- (i) Compute the following tensor product.

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \otimes \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

- (j) How many 1-qubit and 2-qubit gates are needed to perform a 10-qubit Quantum Fourier transform?

Q. 2: Define the following transformation.

$$D = -I + 2|\phi_0\rangle\langle\phi_0|, \text{ where } |\phi_0\rangle = H^{\otimes 3}|0\rangle^{\otimes 3}$$

- (a) Show that $\langle\phi_0|\phi_0\rangle = 1$.
- (b) Using (a), prove that D is unitary.
- (c) Show that $\langle\phi_0|y\rangle = \frac{1}{2\sqrt{2}}$ for all $y \in \{0, 1\}^3$.

Let $\mathcal{S} = \{000, 101, 001, 100\}$. Define the following vectors.

$$|x_{\mathcal{S}}\rangle = \frac{1}{2} \sum_{x \in \mathcal{S}} |x\rangle, \quad |x_{\mathcal{S}}^{\perp}\rangle = \frac{1}{2} \sum_{x \notin \mathcal{S}} |x\rangle$$

- (d) Show that $|x_{\mathcal{S}}\rangle$ and $|x_{\mathcal{S}}^{\perp}\rangle$ are orthonormal vectors.
- (e) Express $|\phi_0\rangle$ as a linear combination of $|x_{\mathcal{S}}\rangle$ and $|x_{\mathcal{S}}^{\perp}\rangle$.
- (f) Using (d) and (e), compute the inner products $\langle\phi_0|x_{\mathcal{S}}\rangle$ and $\langle\phi_0|x_{\mathcal{S}}^{\perp}\rangle$.
- (g) Using (e) and (f), express $D|x_{\mathcal{S}}\rangle$ and $D|x_{\mathcal{S}}^{\perp}\rangle$ as linear combinations of $|x_{\mathcal{S}}\rangle$ and $|x_{\mathcal{S}}^{\perp}\rangle$.
- (h) Consider a state $|\psi\rangle = \frac{1}{2}|x_{\mathcal{S}}\rangle + \frac{\sqrt{3}}{2}|x_{\mathcal{S}}^{\perp}\rangle$. Using (g), express $D|\psi\rangle$ as a linear combination of $|x_{\mathcal{S}}\rangle$ and $|x_{\mathcal{S}}^{\perp}\rangle$.

$$[1 \times 3 + 3 + 1 + 2 \times 2 + 1]$$

Q. 3: An n -qubit Inverse Quantum Fourier Transform (IQFT) is defined as follows.

$$|j\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \exp\left(-\frac{2\pi i j k}{N}\right) |k\rangle, \text{ where } j, k \in \{0, \dots, N-1\}, N = 2^n$$

- (a) Let $\omega = \exp(-2\pi i/N)$. Write the matrix describing the above transform in terms of ω .
- (b) Prove that the above matrix is unitary.
- (c) Show that the above transformation can be equivalently written as follows.

$$|j\rangle \rightarrow \frac{1}{\sqrt{2}} \left(|0\rangle + \exp(-2\pi i 0.j_n) |1\rangle \right) \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + \exp(-2\pi i 0.j_{n-1}j_n) |1\rangle \right) \otimes \dots \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + \exp(-2\pi i 0.j_1 \dots j_n) |1\rangle \right)$$

where $j_1 \dots j_n$ is the binary representation of j .

- (d) Using the above result, construct a quantum circuit for computing 3-qubit IQFT using only 1-qubit and 2-qubit gates. Properly define and label all the gates used.
- (e) Explicitly compute the 3-qubit IQFT of $|100\rangle$ using the result (c).

$$[1 + 2 + 4 + 2 + 1]$$

Q. 4: Consider a point on the Bloch sphere that is characterized by the angles $(\phi, \theta) = (\pi/4, \pi/4)$.

- (a) Write the Bloch coordinates of the point.
- (b) Express the point as an element of the Hilbert space of dimension 2.
- (c) Assume that the state obtained in (b) passes through a σ_Z gate. Write the resulting state as an element of the Hilbert space of dimension 2.
- (d) Compute the Bloch sphere angles (ϕ, θ) of the state computed in (c).

$$[1 \times 4]$$

Q. 5: Consider a 4-qubit composite system whose joint state is described as

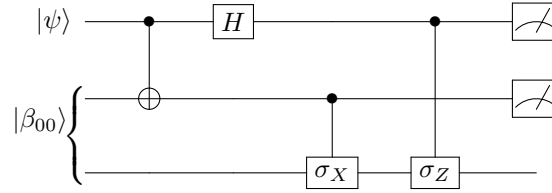
$$|\psi\rangle = \frac{1}{\sqrt{3}} (|0000\rangle + |1011\rangle + |1110\rangle)$$

Assume that the first (leftmost) and the second qubits are jointly measured, and the third and fourth (rightmost) qubits are untouched. Let the ONB for measuring the first qubit is $\{|0\rangle, |1\rangle\}$, and that for measuring the second qubit is $\{|+\rangle, |-\rangle\}$ where

$$|+\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle), \text{ and } |-\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Find out the probabilities of different outcomes and the corresponding joint states after measurements. [8]

Q. 6: Consider the following quantum circuit.



Assume that the initial joint state of the input qubits is $|\psi\rangle \otimes |\beta_{00}\rangle$ where $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ for some complex numbers α, β such that $|\alpha|^2 + |\beta|^2 = 1$, and

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- Compute the joint state after the first and second qubits pass through a CNOT gate.
- Show that the joint state after the first qubit passes through the Hadamard gate can be expressed as:
$$\frac{1}{2}|00\rangle[\alpha|0\rangle + \beta|1\rangle] + \frac{1}{2}|01\rangle[\alpha|1\rangle + \beta|0\rangle] + \frac{1}{2}|10\rangle[\alpha|0\rangle - \beta|1\rangle] + \frac{1}{2}|11\rangle[\alpha|1\rangle - \beta|0\rangle]$$
- Compute the joint state after the second and third qubits pass through the controlled- σ_X gate.
- Obtain the joint state after the first and third qubits pass through the controlled- σ_Z gate.
- Finally, if both first and second qubits are measured using $\{|0\rangle, |1\rangle\}$ as ONBs, what are the possible outcomes and their corresponding probabilities?
- Show that in each possible outcome, the post-measurement state of the third qubit becomes $|\psi\rangle$. Note that $|\psi\rangle$ was originally the state of the first qubit. This phenomenon is typically known as quantum teleportation.

[1 × 6]

Solutions

Q. 1: The answers are provided below.

- (a) All eigenvalues of a unitary matrix have unit modulus. Therefore, $\lambda = 1 + i$ cannot be an eigenvalue of a unitary matrix since $|\lambda| = \sqrt{2}$.
- (b) The states of a 4-qubit composite system are described by a Hilbert space of dimension $2^4 = 16$.
- (c) Number of possible outcomes is $2^3 = 8$.
- (d) Note that $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$. Therefore, $|\psi\rangle$ is not entangled.
- (e) Note that the circuit induces the following basis transformation: $|00\rangle \rightarrow |00\rangle$, $|01\rangle \rightarrow |01\rangle$, $|10\rangle \rightarrow |10\rangle$, $|11\rangle \rightarrow -|11\rangle$. Therefore, the associated unitary matrix is:

$$U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

- (f) Note that, for any $|\phi\rangle, |\psi\rangle$, the following relations hold.

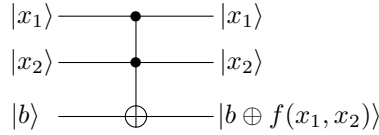
$$\langle (UV)^\dagger \phi | \psi \rangle = \langle \phi | UV \psi \rangle = \langle U^\dagger \phi | V \psi \rangle = \langle V^\dagger U^\dagger \phi | \psi \rangle$$

Therefore, $(UV)^\dagger = V^\dagger U^\dagger$.

- (g) The probability that f will be classified as a constant function is given as:

$$\left| \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \right|^2 = \left| \frac{1}{2^n} \left\{ 3 \cdot 2^{n-2} - 2^{n-2} \right\} \right|^2 = \left(\frac{3}{4} - \frac{1}{4} \right)^2 = \frac{1}{4}$$

- (h) The quantum oracle can be designed using the Toffoli gate as follows.



- (i) The tensor product is given as follows.

$$\begin{pmatrix} 3 & 6 \\ 4 & 8 \end{pmatrix}$$

- (j) Total number of gates needed is $10 + \dots + 1 = 55$.

Q. 2: Note that $|\phi_0\rangle$ can be expressed as follows.

$$\begin{aligned} |\phi_0\rangle &= H^{\otimes 3}|0\rangle^{\otimes 3} = H|0\rangle \otimes H|0\rangle \otimes H|0\rangle \\ &= \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \right]^{\otimes 3} = \frac{1}{2\sqrt{2}} \sum_{x \in \{0,1\}^3} |x\rangle \end{aligned}$$

- (a) The desired inner product can be obtained as follows.

$$\langle \phi_0 | \phi_0 \rangle = \frac{1}{8} \sum_{x \in \{0,1\}^3} \sum_{y \in \{0,1\}^3} \langle x | y \rangle = \frac{1}{8} \sum_{x \in \{0,1\}^3} \sum_{y \in \{0,1\}^3} \mathbf{1}(x = y) = \frac{1}{8} \times 8 = 1$$

where $\mathbf{1}(\cdot)$ is the indicator function.

- (b) The unitarity of D can be shown as follows.

$$\begin{aligned} DD^\dagger &= [-I + 2|\phi_0\rangle\langle\phi_0|] [-I + 2|\phi_0\rangle\langle\phi_0|]^\dagger \\ &= [-I + 2|\phi_0\rangle\langle\phi_0|] [-I + 2|\phi_0\rangle\langle\phi_0|] \\ &= I - 2|\phi_0\rangle\langle\phi_0| - 2|\phi_0\rangle\langle\phi_0| + 4|\phi_0\rangle \underbrace{\langle\phi_0|\phi_0\rangle}_{=1} \langle\phi_0| \\ &= I - 4|\phi_0\rangle\langle\phi_0| + 4|\phi_0\rangle\langle\phi_0| = I \end{aligned}$$

(c) The desired inner product can be obtained as follows.

$$\langle \phi_0 | y \rangle = \frac{1}{2\sqrt{2}} \sum_{x \in \{0,1\}^3} \langle x | y \rangle = \frac{1}{2\sqrt{2}} \sum_{x \in \{0,1\}^3} \mathbf{1}(x = y) = \frac{1}{2\sqrt{2}}$$

(d) Note the following relations.

$$\begin{aligned} \langle x_S | x_S \rangle &= \frac{1}{4} \sum_{x \in S} \sum_{y \in S} \langle x | y \rangle = \frac{1}{4} \sum_{x \in S} \sum_{y \in S} \mathbf{1}(x = y) = \frac{|S|}{4} = 1, \\ \langle x_S^\perp | x_S^\perp \rangle &= \frac{1}{4} \sum_{x \notin S} \sum_{y \notin S} \langle x | y \rangle = \frac{1}{4} \sum_{x \notin S} \sum_{y \notin S} \mathbf{1}(x = y) = \frac{|S^c|}{4} = 1, \\ \langle x_S | x_S^\perp \rangle &= \frac{1}{4} \sum_{x \in S} \sum_{y \notin S} \langle x | y \rangle = \frac{1}{4} \sum_{x \in S} \sum_{y \notin S} \mathbf{1}(x = y) = \frac{|S \cap S^c|}{4} = 0 \end{aligned}$$

(e) The desired linear combination can be obtained as follows.

$$|\phi_0\rangle = \frac{1}{2\sqrt{2}} \sum_{x \in \{0,1\}^3} |x\rangle = \frac{1}{2\sqrt{2}} \left[\sum_{x \in S} |x\rangle + \sum_{x \notin S} |x\rangle \right] = \frac{1}{\sqrt{2}} |x_S\rangle + \frac{1}{\sqrt{2}} |x_S^\perp\rangle$$

(f) The desired inner products can be obtained as follows.

$$\begin{aligned} \langle \phi_0 | x_S \rangle &= \frac{1}{\sqrt{2}} \underbrace{\langle x_S | x_S \rangle}_{=1} + \frac{1}{\sqrt{2}} \underbrace{\langle x_S^\perp | x_S \rangle}_{=0} = \frac{1}{\sqrt{2}}, \\ \langle \phi_0 | x_S^\perp \rangle &= \frac{1}{\sqrt{2}} \underbrace{\langle x_S | x_S^\perp \rangle}_{=0} + \frac{1}{\sqrt{2}} \underbrace{\langle x_S^\perp | x_S^\perp \rangle}_{=1} = \frac{1}{\sqrt{2}} \end{aligned}$$

(g) The desired linear combinations can be obtained as follows.

$$\begin{aligned} D|x_S\rangle &= -|x_S\rangle + 2|\phi_0\rangle\langle\phi_0|x_S\rangle = -|x_S\rangle + \sqrt{2}|\phi_0\rangle \\ &= -|x_S\rangle + \sqrt{2} \left[\frac{1}{\sqrt{2}}|x_S\rangle + \frac{1}{\sqrt{2}}|x_S^\perp\rangle \right] = |x_S^\perp\rangle \end{aligned}$$

In a similar fashion, one can obtain the following.

$$\begin{aligned} D|x_S^\perp\rangle &= -|x_S^\perp\rangle + 2|\phi_0\rangle\langle\phi_0|x_S^\perp\rangle = -|x_S^\perp\rangle + \sqrt{2}|\phi_0\rangle \\ &= -|x_S^\perp\rangle + \sqrt{2} \left[\frac{1}{\sqrt{2}}|x_S\rangle + \frac{1}{\sqrt{2}}|x_S^\perp\rangle \right] = |x_S\rangle \end{aligned}$$

(h) The desired result is stated below.

$$D|\psi\rangle = \frac{1}{2}D|x_S\rangle + \frac{\sqrt{3}}{2}D|x_S^\perp\rangle = \frac{1}{2}|x_S^\perp\rangle + \frac{\sqrt{3}}{2}|x_S\rangle$$

Q. 3: The answers are stated below.

(a) The required matrix can be expressed as

$$\frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{(N-1)} & \omega^{2(N-1)} & \dots & \omega^{(N-1)^2} \end{pmatrix}$$

(b) Let $C_k = \frac{1}{\sqrt{N}}[1, \omega^k, \omega^{2k}, \dots, \omega^{(N-1)k}]^\top$, $k \in \{0, 1, \dots, N-1\}$ be the k th column of the above matrix. To prove that IQFT is a unitary transformation, it is sufficient to show that $C_j^\dagger C_k = \delta_{jk}$ for any $j, k \in \{0, 1, \dots, N-1\}$. This result is proven below. Note that

$$C_j^\dagger C_k = \frac{1}{N} \sum_{l=0}^{N-1} \bar{\omega}^{jl} \omega^{kl} = \frac{1}{N} \sum_{l=0}^{N-1} \omega^{(k-j)l}$$

If $j = k$, we obtain

$$C_j^\dagger C_j = \frac{1}{N} \sum_{l=0}^{N-1} 1 = 1$$

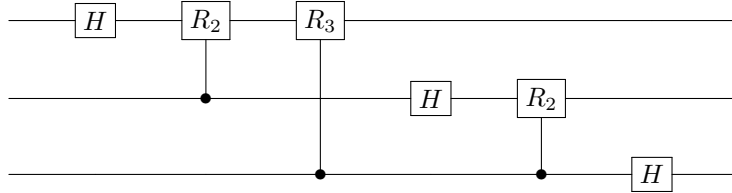
If $j \neq k$, we obtain,

$$C_j^\dagger C_k = \frac{1}{N} \sum_{l=0}^{N-1} [\omega^{(k-j)}]^l = \frac{1}{N} \frac{1 - \omega^{N(k-j)}}{1 - \omega^{(k-j)}} = \frac{1 - [\exp(2\pi i(k-j)/N)]}{N [1 - \exp(2\pi i(k-j)/N)]} = 0$$

(c) The desired result can be proven as follows.

$$\begin{aligned} |j\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \exp\left(-\frac{2\pi i j k}{N}\right) |k\rangle \\ &= \frac{1}{2^{n/2}} \sum_{k_1=0}^1 \cdots \sum_{k_n=0}^1 \exp\left(-2\pi i j \sum_{l=1}^n k_l 2^{-l}\right) |k_1 k_2 \cdots k_n\rangle \\ &= \bigotimes_{l=1}^n \left[\frac{1}{\sqrt{2}} \sum_{k_l=0}^1 \exp(-2\pi i j k_l 2^{-l}) |k_l\rangle \right] \\ &= \bigotimes_{l=1}^n \frac{1}{\sqrt{2}} \left[|0\rangle + \exp(-2\pi i j 2^{-l}) |1\rangle \right] \\ &= \frac{1}{\sqrt{2}} \left[|0\rangle + \exp(-2\pi i 0.j_n) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp(-2\pi i 0.j_{n-1}j_n) |1\rangle \right] \otimes \\ &\quad \cdots \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp(-2\pi i 0.j_1 \cdots j_n) |1\rangle \right] \end{aligned}$$

(d) The circuit for implementing a 3-qubit IQFT is shown below.



where the gates R_k , $k \in \{2, 3\}$ are defined below:

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & \exp\left(-\frac{2\pi i}{2^k}\right) \end{bmatrix}$$

(e) The explicit 3-qubit IQFT of $|100\rangle$ is given as:

$$\begin{aligned} &\frac{1}{\sqrt{2}} \left[|0\rangle + \exp(-2\pi i 0.0) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp(-2\pi i 0.00) |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + \exp(-2\pi i 0.100) |1\rangle \right] \\ &= \frac{1}{\sqrt{2}} \left[|0\rangle + |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle + |1\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle - |1\rangle \right] \\ &= \frac{1}{2} \left[|00\rangle + |01\rangle + |10\rangle + |11\rangle \right] \otimes \frac{1}{\sqrt{2}} \left[|0\rangle - |1\rangle \right] \\ &= \frac{1}{2\sqrt{2}} \left[|000\rangle + |010\rangle + |100\rangle + |110\rangle - |001\rangle - |011\rangle - |101\rangle - |111\rangle \right] \end{aligned}$$

Q. 4: Recall that the Bloch coordinates of a point on the Bloch sphere with angles (ϕ, θ) are given as

$$(\cos \phi \sin \theta, \sin \phi \sin \theta, \cos \theta)$$

Moreover, the above point is represented by the following vector in the Hilbert space

$$\cos\left(\frac{\theta}{2}\right) |0\rangle + \exp(i\phi) \sin\left(\frac{\theta}{2}\right) |1\rangle$$

(a) The desired Bloch coordinates are given below

$$\left(\cos\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{4}\right), \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{4}\right), \cos\left(\frac{\pi}{4}\right) \right) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right)$$

(b) The Hilbert space representation of the above point is

$$\cos\left(\frac{\pi}{8}\right) |0\rangle + \exp\left(\frac{i\pi}{4}\right) \sin\left(\frac{\pi}{8}\right) |1\rangle$$

(c) After passing through the σ_Z , the state changes to

$$\cos\left(\frac{\pi}{8}\right) |0\rangle - \exp\left(\frac{i\pi}{4}\right) \sin\left(\frac{\pi}{8}\right) |1\rangle = \cos\left(\frac{\pi}{8}\right) |0\rangle + \exp\left(\frac{5\pi i}{4}\right) \sin\left(\frac{\pi}{8}\right) |1\rangle$$

(d) Hence, the Bloch sphere angles of the above point is $(\pi, \theta) = (5\pi/4, \pi/4)$.

Q. 5: There are four possible outcomes, namely $\{(0, +), (0, -), (1, +), (1, -)\}$.

Case I: The probability of $(0, +)$ outcomes is

$$P_{0+} = \|\langle 0|\langle +|\psi\rangle\|^2 = \left\| \frac{1}{\sqrt{2}} (\langle 00| + \langle 01|) |\psi\rangle \right\|^2 = \left\| \frac{1}{\sqrt{6}} |00\rangle \right\|^2 = \frac{1}{6}$$

Moreover, the post-measurement joint state is

$$|0\rangle|+\rangle \otimes \frac{\langle 0|\langle +|\psi\rangle}{\|\langle 0|\langle +|\psi\rangle\|} = |0\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |00\rangle = \frac{1}{\sqrt{2}} [|0000\rangle + |0100\rangle]$$

Case II: The probability of $(0, -)$ outcomes is

$$P_{0-} = \|\langle 0|\langle -|\psi\rangle\|^2 = \left\| \frac{1}{\sqrt{2}} (\langle 00| - \langle 01|) |\psi\rangle \right\|^2 = \left\| \frac{1}{\sqrt{6}} |00\rangle \right\|^2 = \frac{1}{6}$$

Moreover, the post-measurement joint state is

$$|0\rangle|-\rangle \otimes \frac{\langle 0|\langle -|\psi\rangle}{\|\langle 0|\langle -|\psi\rangle\|} = |0\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes |00\rangle = \frac{1}{\sqrt{2}} [|0000\rangle - |0100\rangle]$$

Case III: The probability of $(1, +)$ outcomes is

$$P_{1+} = \|\langle 1|\langle +|\psi\rangle\|^2 = \left\| \frac{1}{\sqrt{2}} (\langle 10| + \langle 11|) |\psi\rangle \right\|^2 = \left\| \frac{1}{\sqrt{6}} (|11\rangle + |10\rangle) \right\|^2 = \frac{1}{3}$$

Moreover, the post-measurement joint state is

$$\begin{aligned} |1\rangle|+\rangle \otimes \frac{\langle 1|\langle +|\psi\rangle}{\|\langle 1|\langle +|\psi\rangle\|} &= |1\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|11\rangle + |10\rangle) \\ &= \frac{1}{2} [|1011\rangle + |1010\rangle + |1111\rangle + |1110\rangle] \end{aligned}$$

Case IV: The probability of $(1, -)$ outcomes is

$$P_{1-} = \|\langle 1|\langle -|\psi\rangle\|^2 = \left\| \frac{1}{\sqrt{2}} (\langle 10| - \langle 11|) |\psi\rangle \right\|^2 = \left\| \frac{1}{\sqrt{6}} (|11\rangle - |10\rangle) \right\|^2 = \frac{1}{3}$$

Moreover, the post-measurement joint state is

$$\begin{aligned} |1\rangle|-\rangle \otimes \frac{\langle 1|\langle -|\psi\rangle}{\|\langle 1|\langle -|\psi\rangle\|} &= |1\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}} (|11\rangle - |10\rangle) \\ &= \frac{1}{2} [|1011\rangle - |1010\rangle - |1111\rangle + |1110\rangle] \end{aligned}$$

Q. 6: Note that the initial joint state is

$$|\psi\rangle \otimes |\beta_{00}\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = \frac{\alpha}{\sqrt{2}} |0\rangle [|00\rangle + |11\rangle] + \frac{\beta}{\sqrt{2}} |1\rangle [|00\rangle + |11\rangle]$$

(a) After the first CNOT gate, the joint state changes to

$$\frac{\alpha}{\sqrt{2}}|0\rangle\left[|00\rangle+|11\rangle\right]+\frac{\beta}{\sqrt{2}}|1\rangle\left[|10\rangle+|01\rangle\right]$$

(b) After the Hadamard gate, the joint state changes to:

$$\begin{aligned} & \frac{\alpha}{2}\left[|0\rangle+|1\rangle\right]\otimes\left[|00\rangle+|11\rangle\right]+\frac{\beta}{2}\left[|0\rangle-|1\rangle\right]\otimes\left[|10\rangle+|01\rangle\right] \\ &= \frac{1}{2}|00\rangle\left[\alpha|0\rangle+\beta|1\rangle\right]+\frac{1}{2}|01\rangle\left[\alpha|1\rangle+\beta|0\rangle\right]+\frac{1}{2}|10\rangle\left[\alpha|0\rangle-\beta|1\rangle\right]+\frac{1}{2}|11\rangle\left[\alpha|1\rangle-\beta|0\rangle\right] \end{aligned}$$

(c) After the controlled- σ_X gate, the joint state changes to:

$$\frac{1}{2}|00\rangle\left[\alpha|0\rangle+\beta|1\rangle\right]+\frac{1}{2}|01\rangle\left[\alpha|0\rangle+\beta|1\rangle\right]+\frac{1}{2}|10\rangle\left[\alpha|0\rangle-\beta|1\rangle\right]+\frac{1}{2}|11\rangle\left[\alpha|0\rangle-\beta|1\rangle\right]$$

(d) After the controlled- σ_Z gate, the joint state finally changes to:

$$\begin{aligned} |\psi_f\rangle &= \frac{1}{2}|00\rangle\left[\alpha|0\rangle+\beta|1\rangle\right]+\frac{1}{2}|01\rangle\left[\alpha|0\rangle+\beta|1\rangle\right]+\frac{1}{2}|10\rangle\left[\alpha|0\rangle+\beta|1\rangle\right]+\frac{1}{2}|11\rangle\left[\alpha|0\rangle+\beta|1\rangle\right] \\ &= \frac{1}{2}\left[|00\rangle+|01\rangle+|10\rangle+|11\rangle\right]\otimes\left[\alpha|0\rangle+\beta|1\rangle\right] \end{aligned}$$

(e) There are four possible outcomes, namely $\{00, 01, 10, 11\}$. The probability of the outcome jk where $j, k \in \{0, 1\}$ is

$$P_{jk} = \|\langle jk|\psi_f\rangle\|^2 = \left\|\frac{1}{2}[\alpha|0\rangle+\beta|1\rangle]\right\|^2 = \frac{1}{4} [|\alpha|^2+|\beta|^2] = \frac{1}{4}$$

(f) Moreover, the post-measurement state is

$$|jk\rangle\otimes\frac{\langle jk|\psi_f\rangle}{\|\langle jk|\psi_f\rangle\|} = |jk\rangle\otimes\left[\alpha|0\rangle+\beta|1\rangle\right]$$

Hence, for each possible outcome, the post-measurement state of the third qubit is $\alpha|0\rangle+\beta|1\rangle = |\psi\rangle$.